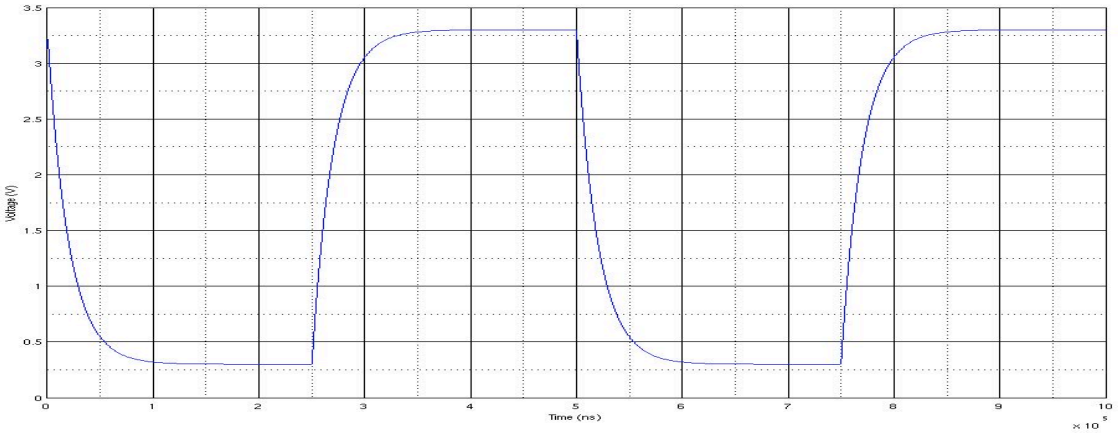
Final Exam. #2 VLSI. Spring 2012. Name _____

The sheet resistance is 10^4 ohms per square for a turned-on NMOS and use double that value for PMOS. Cg=10^{-14} Farads for a transistor whose size is 2 units by 2 units. Cg is the capacitance of a 2 x 2 NMOS or PMOS gate. Use VT = 1 V for nmos and –1V for pmos. Unless stated otherwise, assume Vdd=5V. Note: variables are given by single characters, e.g. A, B, …, and concatenated character strings such as AB are to be read as A and B. A' means Acomplement. A + B means A or B.



1 2 3 4 5 6 7 8 9 10

1. _____ *[5] What is the apparent pullup resistance of the pullup transistor and the apparent pulldown resistance of pulldown transistor if this is the output of an inverter that drives a 1000 pF load?*

   Rpullup=_____Rpulldown=_____

2. *[5] Explain self loading of a transistor by its own drain.*

3. *[4]Draw the input output curve demonstrating the characteristics of a pseudo_nmos inverter. Clearly indicate Vlo and Vhi. The ordinate and the abscissa of your graph should go from 0 to Vdd.*

4. ***[6] What is the input capacitance of the inverter sketched below? Take the Miller effect into account, utilizing the physical circuit element Cm which is drawn in the circuit. Give your answer as a mathematical expression that includes Cg and Cm as parameters.***

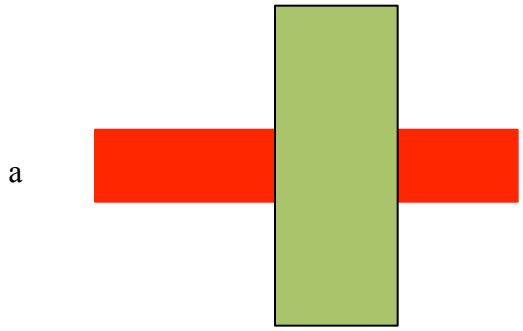
5. ***[5]To a first approximation, what is the input capacitance seen looking into the inverter shown below. The load is 800fF. Ignore the Miller Effect.***

6. ***[5]Sheet resistance. A poly wire of dimensions 2 x 1600 has a sheet resistance of 200 ohms per square. What is its resistance? Assume the current goes through the wire via the "long" dimension.***

d

c

7. *[5] Extract (draw) a circuit model. The nodes you will use are {a,c,d}. Make up your own labels for the extracted elements. If you drew this on a screen, you would do it using TWO rectangles. Red is poly. Green is n-diffusion.*



8. *[10] The tau model. A 1x fully-complementary CMOS inverter drives an identical 10x inverter. (A 10x inverter means that the widths of pull-ups and pulldown transistors are of width 20 for the NMOS transistor and 40 for the PMOS transistor. The gate length is always 2.) What is the pulldown time at the output of the first inverter? Use Tau_nmos_2= 0.1ns. Use 0.2ns for the PMOS transistor.*

9. *[10] The RC model. A 1x fully-complementary CMOS inverter drives a 100fF load. Assume the NMOS transistor has a sheet resistance of 10kohm and PMOS transistors have 20kohm, what is the pullup time constant?*

The sheet resistance is 10^4 ohms per square for a turned-on NMOS and use double that value for PMOS. Cg=10^{-14} Farads for a transistor whose size is 2 units by 2 units. Cg is the capacitance of a 2 x 2 NMOS or PMOS gate. Use V_T = 1 V for nmos and –1V for pmos. Unless stated otherwise, assume Vdd=5V. Note: variables are given by single characters, e.g. A, B, …, and concatenated character strings such as AB are to be read as A and B. A' means Acomplement. A + B means A or B.

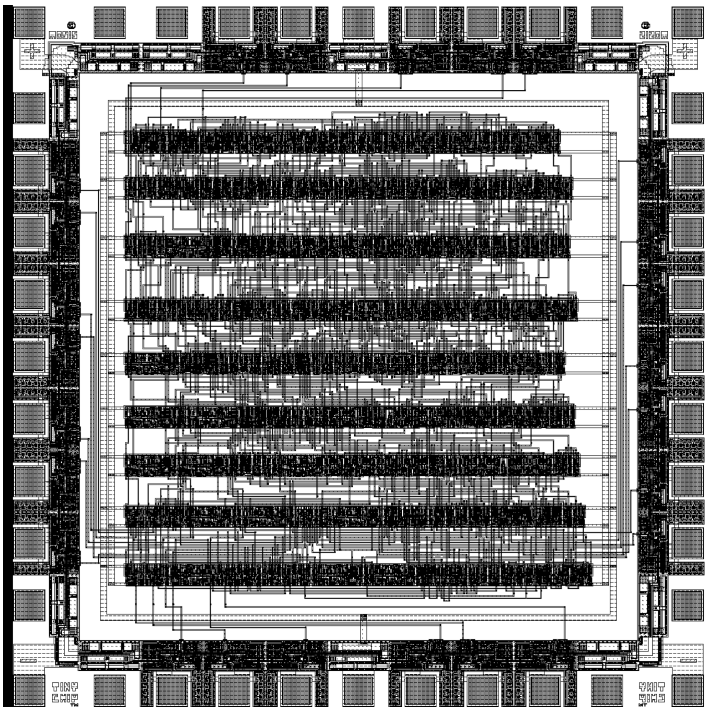
10. *(6)Design NAND(A,B) such that the pullup time is approximately 1 ns when A=0 and B=1. Let the load be 100fF. Draw the circuit showing transistors and labels A and B, but only show sizes for the relevant transistor or transistors.*

11. ***(6) Realize (means draw the circuit) the function F=(AB+C) using a fully-complementary, fully switched (as contrasted with ratioed) CMOS circuit. Show the transistor level schematic. The signal, F, should be true at the output of the circuit. Use a design style that DOES require the use of an inverter, so that F appears true at the output.***

12. ***[9])Realize the function F=AB+CD using three domino stages. Indicate the precharge and evaluate signals. Be sure the output is a "valid" domino signal that can be fed to other NMOS logic, and that signal should be "true", not "false".***

13. ***[4]Draw a transistor schematic of a quasi-static DFlipFlop. Label clocks phi1 and phi2. Label Din and Dout.***

14. ***Design methodologies. [4] Give the style a name._____. About how large in microns on a side do you estimate this chip to be?***

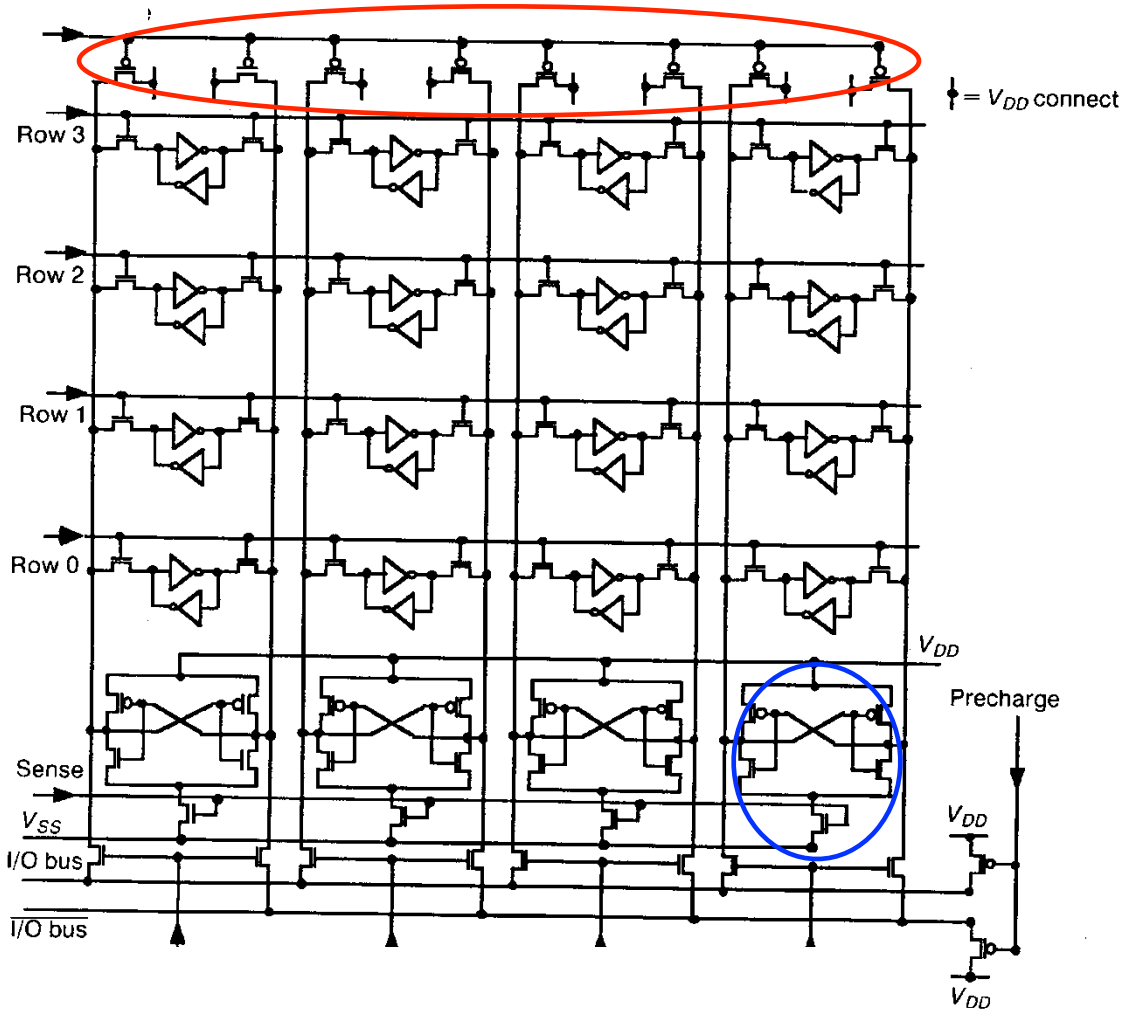


15. [6]For the schematic shown below there are several questions

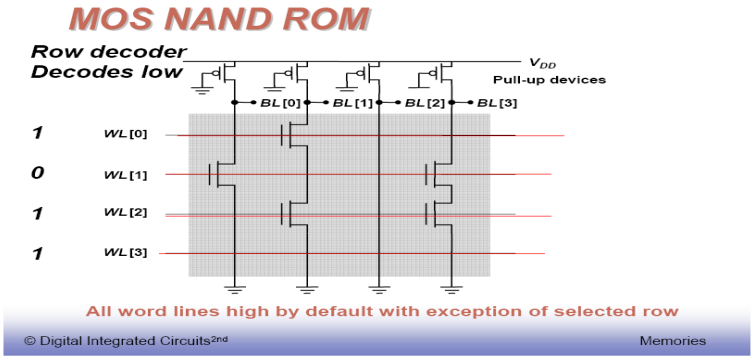
a. What is the overall structure?_____.

b. What is the purpose of the transistors shown in red?

c. What is the structure circled in BLUE?



16. *[4]What are the contents of address {01}? The answer fits in FOUR spaces.*_____



17. *[6]What is LSSD testing of a VLSI chip? Include in your explanation the words "serial", "Flipflops"or "Latches", and "chained together", and "observability" and "Controlability.*